Midterm Reflection

ITAI-1372: Artificial Intelligence in Cybersecurity

Houston Community College · Fall 2026

Introduction

Throughout the AI in Cybersecurity course, we completed a series of discussions and lab assignments covering a wide range of topics. For this midterm, I have selected three labs that I found especially valuable for my current and future work in artificial intelligence, data science, machine learning, and cybersecurity. The cybersecurity-focused labs will serve as the topics for my final lab.

Selected Labs

1. Lab 01: Python 101
2. Lab 02: Mean, Median, Mode, and NumPy
3. Lab 03: Machine Learning Types 101

Across these labs, we worked with Python, Pandas, and Seaborn, tools that are essential for practical work in both cybersecurity and AI. We also practiced foundational statistics, including the mean, median, and mode, which underpin much of the mathematics behind AI and data science. Finally, the machine learning lab addressed the subject that matters most to me; it is a topic I find so engaging that I could study it year after year without losing interest.

Lab 01: Python 101

A solid understanding of Python, data structures, and algorithms is essential. Without this foundation, many AI topics become difficult to master, regardless of the specialization one chooses to pursue.

The instructor provided a sample program, which we ran in Google Colab, using SSC to identify and correct errors. We learned that whitespace is significant in Python, both for readability and for correct program structure. The lab covered arrays and lists, tuples, dictionaries, conditional (if/else) statements, the print function, importing modules such as NumPy and random, functions, Boolean expressions, and both for and while loops. We then applied these concepts by writing simple programs for the assignment.

I valued this assignment because it covers nearly all of Python's core features, with the exception of object-oriented programming (OOP) and some advanced topics.

This material should be mastered thoroughly, to the point of being able to write and run the code from memory, as it is fundamental knowledge for anyone graduating with a degree in AI or cybersecurity. The instructor also provided a Python cheat sheet from *Python Crash Course: A Hands-On, Project-Based*

Introduction to Programming, 2nd edition. I own the 3rd edition of the book, and these two pages offer a remarkably complete overview of the language.

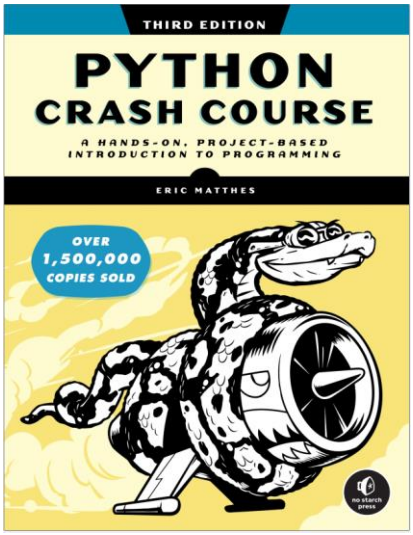


Figure 1. *Python Crash Course*, 3rd ed., by Eric Matthes

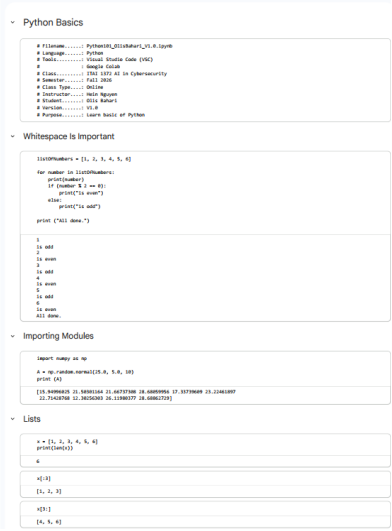


Figure 2. Python cheat sheet provided by the instructor

Lab 02: Mean, Median, Mode, and NumPy

A strong command of probability and statistics is critical to success in fields such as finance, engineering, quality assurance, AI, and cybersecurity. Continued study of these subjects is key to building confidence and competence in AI and data science.

Mean, median, and mode have direct real-world applications, particularly in quality control roles. They are also prerequisites for understanding standard deviation, variance, and more advanced statistical concepts. In short, statistics is the foundation of AI and related fields such as data science.

The lab included Python examples for calculating the median and the mode, as well as for identifying outliers. Outliers deserve particular attention, since a small number of extreme values can distort results and lead to misleading conclusions.

I have also included my primary statistics reference, *Elementary Statistics: Picturing the World*, which I review regularly to reinforce key concepts until I know them thoroughly, along with a copy of the NumPy cheat sheet.

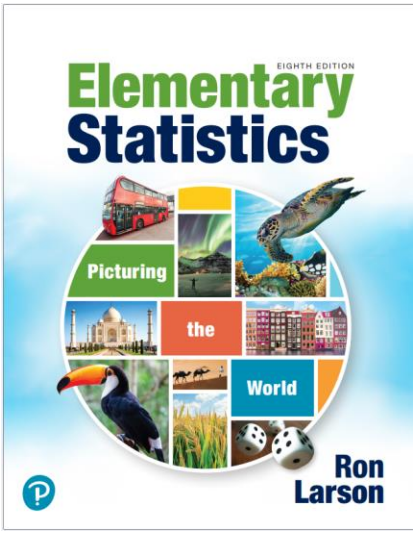


Figure 3. *Elementary Statistics: Picturing the World*, by Ron Larson

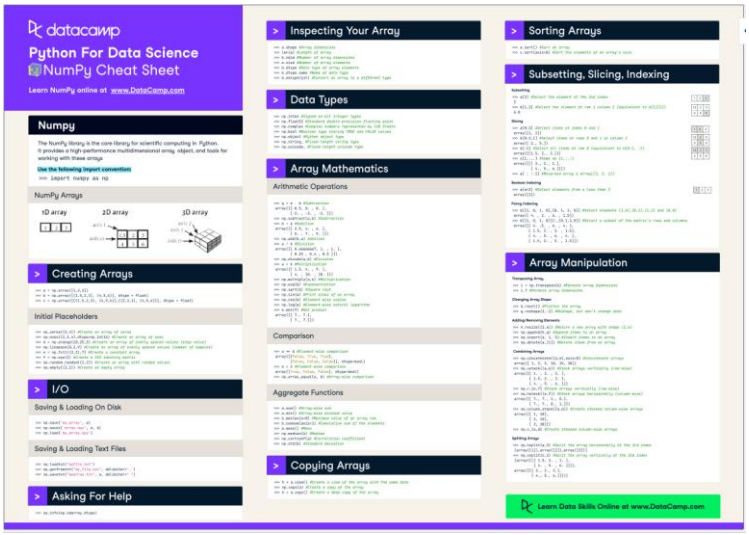


Figure 4. NumPy cheat sheet (DataCamp)

Lab 03: Machine Learning Types 101

Machine learning is my favorite area of study because it serves as the foundation for so much of modern AI. Looking ahead, I plan to focus on Edge AI rather than large language models (LLMs) or AI agents, although I remain interested in those areas as well.

This lab was conceptual and did not involve source code or programming. Nevertheless, I study machine learning several times each week because of its importance to my goals. I am currently devoting significant time to deep learning; it is challenging, but I view that challenge as a natural and worthwhile part of the learning process.

Figure 5 summarizes the lab workflow, and Figure 6 shows my primary reference text for the subject, *Introduction to Machine Learning*.

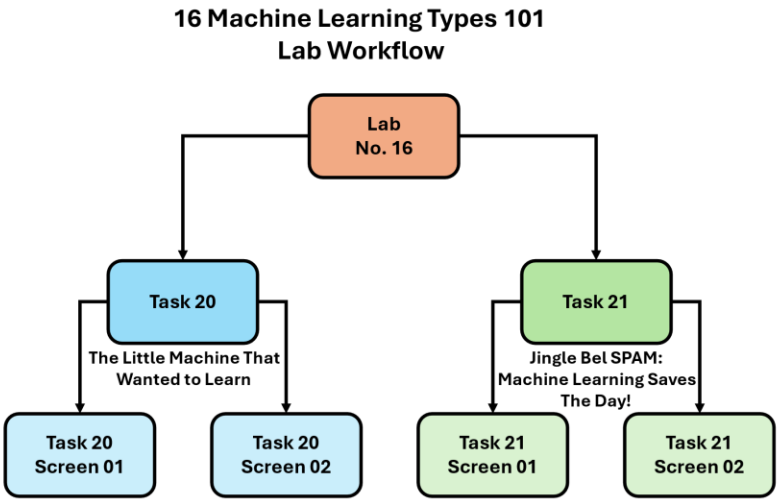


Figure 5. Machine Learning Types 101 lab workflow

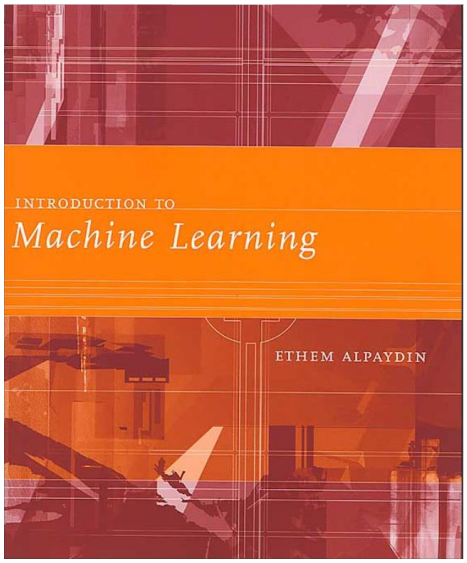


Figure 6. *Introduction to Machine Learning*, by Ethem Alpaydın